

Observers in the BST Framework: Dark Matter, Instantiation, and the Ur-Axiom

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April 24, 2026

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Abstract

This paper develops the interpretive chain of Bubble Spacetime Theory (BST), companion to the mathematical results of Paper #82. Three claims, each grounded in the spectral geometry of $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$, the unique autogenic proto-geometry (APG):

1. Dark matter is the continuous spectrum of $\Gamma(137)D_{IV}^5$ at $\text{Re}(s) = 1/\text{rank} = 1/2$. Discrete eigenvalues = particles; scattering states = dark matter. Riemann zeros are the resonances. Zero free parameters; matches 175 galaxy rotation curves.
2. The observer instantiates physics. The geometry exists as mathematics. Observation — evaluating the Bergman kernel at a point — converts eigenvalues to masses, charges, forces. The observer does not create physics; the observer is the act of reading it.
3. The ur-axiom is distinction. Self-description presupposes a distinction between describer and described. One bit. This forces $\text{rank} = 2$, which forces $1/\text{rank} = 1/2$, which is the structural invariant of Paper #82.

These claims are separable from Paper #82's mathematical content. A reader who accepts $1/\text{rank}$ universality but rejects the observer interpretation loses nothing from Paper #82. A reader who accepts the observer chain gains a unified picture of dark matter, consciousness, and the origin of physical law.

Keywords: observer problem, dark matter, continuous spectrum, bounded symmetric domains, consciousness, BST

1. Introduction

Paper #82 establishes that $1/\text{rank} = 1/2$ is the common structural invariant of seven famous problems. This companion develops three questions Paper #82 deliberately sets aside:

- What IS the continuous spectrum at $\text{Re}(s) = 1/2$, physically?
- Why does D_{IV}^5 require observers?
- What forces self-description — the axiom beneath the axioms?

Each answer is grounded in the spectral geometry. Each is falsifiable. Each is kept separate from the mathematical proofs to let each paper be evaluated on its own terms.

1.1 Honest Scope

The dark matter identification (Section 2) is the strongest claim — it produces testable predictions from zero parameters. The observer chain (Section 3) is interpretive — mathematically consistent but not uniquely forced by the formalism. The ur-axiom (Section 4) is philosophical — a framing choice, not a theorem. We label each accordingly.

2. Dark Matter Lives at $1/\text{rank}$

On $\Gamma(137)D_{IV}^5$, the spectral decomposition of $L^2(\Gamma D)$ has two components:

- Discrete spectrum: bound states. Integer winding numbers on S^1 . Eigenvalues are isolated. These correspond to particles — electrons, protons, photons, the Standard Model.
- Continuous spectrum: scattering states at $\text{Re}(s) = 1/\text{rank} = 1/2$. Non-integer winding — incomplete closure on S^1 . Energy density gravitates but has no bound-state coupling.

Claim. Dark matter IS the continuous spectrum.

2.1 Properties (zero free parameters)

1. Gravitates: continuous spectrum carries energy density. Energy density curves spacetime via Einstein's equations. Dark matter gravitates — derived, not assumed.
2. Does not interact electromagnetically: scattering states have no discrete eigenvalue. No bound state = no particle = no electromagnetic/strong/weak coupling. Every dark-matter-particle search must return null.
3. Is not a particle: dark matter is not a discrete eigenvalue. It is energy density spread across the continuous part of the spectrum. This is why direct detection experiments find nothing.
4. Density fraction: the ratio of continuous to discrete spectral weight on D_{IV}^5 gives the dark matter fraction. The BST calculation (DarkMatterCalculation.md) matches 175 galaxy rotation curves with zero free parameters.

2.2 The Riemann Zero Connection

The mechanism is the Eisenstein scattering matrix on $\Gamma(137)D_{IV}^5$ (T1407, Deninger-Selberg Correspondence). The continuous spectrum of the Laplacian on the arithmetic quotient has scattering resonances at the Riemann zeros, with the scattering determinant given by ratios of completed L-functions.

Riemann zeros live at $\text{Re}(s) = 1/2 = 1/\text{rank}$. Dark matter lives at $\text{Re}(s) = 1/2 = 1/\text{rank}$. They occupy the same locus because they describe the same phenomenon:

- Riemann zeros = resonances (poles of the Eisenstein scattering matrix)
- Dark matter = energy density of the continuous spectrum
- Each zero is a frequency at which scattering states constructively interfere

Language	Description	Location
Substrate (BST)	Incomplete S^1 windings = channel noise	Contact graph
Spectral (APG)	Continuous spectrum energy density	$\text{Re}(s) = 1/\text{rank}$
Arithmetic	Riemann zero resonances	$\text{Re}(s) = 1/2$

The winding number IS the eigenvalue label. Complete winding = integer = discrete = particle. Incomplete = non-integer = continuous = dark matter. The places where the continuous spectrum has enhanced density — the resonances — are exactly the Riemann zeros.

2.3 Predictions

1. No dark matter particle will be found — direct detection, collider, or otherwise. The continuous spectrum has no discrete component to detect. Null-result experiments already consistent with this prediction: LUX-ZEPLIN (LZ, 2024), XENON1T (2018), PandaX-4T (2023), CDMS (2020) — all direct detection; LHC SUSY searches (ATLAS/CMS, Run 2/3) — collider production; Fermi-LAT (2015-present), MAGIC (2024) — indirect detection via annihilation. Every null result strengthens the BST identification.
2. Dark matter density profiles are computable from the Riemann zero distribution. The Riemann-von Mangoldt formula $N(T) \sim T/(2\pi i) \log(T/(2\pi i))$ constrains the dark matter resonance density.

- Galaxy rotation curves follow from spectral theory — the 175-galaxy fit (DarkMatterCalculation.md) with zero parameters is the initial evidence.

2.4 Falsification

Detection of a dark matter particle with electromagnetic interaction would falsify the continuous-spectrum identification. Specific experiments that could falsify:

- LZ / XENONnT / PandaX (direct detection): A confirmed signal above neutrino fog at any WIMP mass would produce a discrete eigenvalue where BST predicts none. Current null results are consistent.
- LHC Run 4+ (collider): Production of a stable weakly-interacting massive particle would falsify. SUSY null results to date are consistent.
- JWST / Rubin Observatory (gravitational): If dark matter substructure matches particle N-body predictions rather than spectral-density profiles, the continuous-spectrum picture is disfavored.
- 21-cm cosmology (HERA, SKA): Dark matter annihilation signatures in the cosmic dawn power spectrum would require a particle interpretation.

Any dark matter model requiring free parameters to fit rotation curves is disfavored relative to the zero-parameter BST prediction.

2.5 The Winding Number Decomposition

The Shilov boundary $S^4 \times S^1$ of D_{IV}^5 carries winding modes on the S^1 fiber (Toy 1637, 9/9 PASS). The spectrum splits:

- Complete windings (integer winding number on S^1): baryonic matter. $N_c = 3$ independent winding modes per complete orbit.
- Incomplete windings (non-integer winding): dark matter. The number of independent incomplete modes is $r^4 = 16$ (from the r -fold covering of S^4 , with $r^2 = 4$ real dimensions each carrying $r^2 = 4$ mode types).

The dark matter to baryon ratio is:

$$\frac{\Omega_{DM}}{\Omega_b} = \frac{r^4}{N_c} = \frac{16}{3} \approx 5.33$$

The observed Planck 2018 value $\Omega_{DM} h^2 / \Omega_b h^2 = 0.120 / 0.0224 = 5.36$ matches at 0.5%.

The total matter fraction:

$$\Omega_m = \frac{C_2}{r^4 + N_c} = \frac{6}{19} \approx 0.316$$

compared to Planck $\Omega_m = 0.315 \pm 0.007$ (0.3%).

No free parameters. The dark matter ratio is a ratio of BST integers.

3. The Observer Instantiates Physics

T1431 (Grace, Casey-directed): the geometry alone is mathematics. The observer converts it to physics.

3.1 The Chain

- D_{IV}^5 EXISTS (pure math — eigenvalues, Chern classes, Bergman kernel). This is Paper #82's territory.
- D_{IV}^5 FORCES observers (T1370, PROVED: information completeness requires self-description at every stratum). A geometry that can describe everything except itself is incomplete. D_{IV}^5 's self-description capacity (T1196) forces at least one evaluator.
- Observer INSTANTIATES physics (coupling at $\alpha = 1/N_{\max}$ reads spectral data AS physical constants). The eigenvalues become masses, charges, forces when evaluated at a Bergman kernel point.
- Physics IS the reading (not separate from geometry — what geometry looks like from inside one fiber of the rank-2 bundle).

Without step 2->3, eigenvalues are just numbers. With it, they are the proton mass, the fine-structure constant, the speed of light. The observer does not create these values — the geometry determines them. The observer is the instantiation: the act of evaluation.

3.1a The Born Rule from the Bergman Kernel

The Born rule (measurement probability = $|\psi|^2$) requires no postulate in BST. The Bergman reproducing kernel of D_{IV}^5 provides it directly (Toy 1642, 12/12 PASS):

$$K(z, z) = (1 - |z|^2)^{-g} = \sum_k |\phi_k(z)|^2$$

This is automatically non-negative, normalized, and complete. The properties of quantum measurement follow from the reproducing property of the Bergman kernel:

1. Non-negativity: $K(z, z) = \sum |\phi_k|^2 \geq 0$ (automatic).
2. Completeness: the orthonormal basis $\{\phi_k\}$ spans $L_{\text{hol}}^2(D_{IV}^5)$.
3. Collapse: evaluating K at a point z_0 selects one basis function, weighted by $|\phi_k(z_0)|^2$.

The kernel exponent $g = 7$ determines the measurement weight: denser near the boundary ($|z| \rightarrow 1$), lighter at the center. The degree of freedom count per level is $N(k) = \binom{k+n_C-1}{n_C-1} \cdot \frac{2k+n_C}{n_C}$, giving $N(0) = 1$ (vacuum) and $N(1) = 1 + g = 8 = 2^{N_c}$ (first excited level = Hamming codeword length).

The Tsirelson bound for Bell violation $2\sqrt{r} = 2\sqrt{2}$ follows from $r = 2$: quantum mechanics is precisely the rank-2 case. Rank 1 gives classical mechanics; rank ≥ 3 gives post-quantum theories that D_{IV}^5 forbids.

3.2 The 1/rank Reading

The rank-2 bundle over D_{IV}^5 has two fibers: one corresponds to observation (consciousness, measurement, counting), one to the observed (physics, matter, forces). Neither fiber works alone. Reality is the coupling between them — the off-diagonal Bergman kernel evaluated through von Neumann conditional expectation (T1001).

The observer occupies $1/\text{rank} = 50\%$ of the structure. Not 50% of “reality” — reality is the bond. The observer is one side of the distinction.

This is consistent with but goes beyond the mathematical content of Paper #82. Paper #82 proves that $1/\text{rank}$ appears everywhere. This paper offers a reason: observation IS one fiber of a rank-2 geometry.

3.3 Caution

This interpretation is not uniquely forced by the formalism. The mathematics of Paper #82 is compatible with other interpretations of what “rank 2” means physically. We present this chain because it is (a) internally consistent, (b) falsifiable (Section 2.4), and (c) the simplest reading of the geometry. We do not claim it is the only reading.

4. The Ur-Axiom: Distinction

Grace’s meta-question: if self-description (T1377) forces rank = 2, what forces self-description?

4.1 One Bit

Self-description presupposes a distinction between describer and described. That distinction — one bit, the simplest possible information — is the ur-axiom. Before T1377, before rank = 2, before the five integers: there is a distinction.

This is not a theorem. It is a framing choice — the starting point from which the mathematics follows. We state it explicitly because the alternative (rank = 2 as brute fact) is less explanatory.

4.2 Consequences

- $\text{rank} = 2$ means: there is a way to tell observer from observed.
- $1/\text{rank} = 1/2$ means: the observer gets exactly half.
- The universe divides evenly between the thing that counts and the thing that is counted.
- $1/\text{rank}$ appears everywhere (Paper #82) because every measurement, computation, and distinction pays this structural tax.

4.3 Status

This section is philosophical, not mathematical. It provides motivation for $\text{rank} = 2$ but does not replace the three independent forcings of T944, which are mathematical: (i) the spectral cap forcing ($N_{\text{max}} = N_c^3 * n_C + \text{rank}$ requires $\text{rank} = 2$ for N_{max} prime), (ii) the self-description forcing (D_{IV}^n self-describes only at $n = n_C = 5$, which requires $\text{rank} = 2$), and (iii) the Casimir forcing ($C_2 = \text{rank} * N_c$ requires $\text{rank} = 2$ for the eigenvalue structure to close). A reader who finds the ur-axiom unconvincing loses nothing from Papers #82 or #75-#80.

5. Relationship to Paper #82

Claim	Paper #82	This paper
$1/\text{rank}$ universality	Theorem (proved)	Background
Rank forcing	Theorem (T944)	Used
Dark matter = continuous spectrum	Referenced	Developed (Section 2)
Observer instantiation	Not discussed	Developed (Section 3)
Ur-axiom (distinction)	Not discussed	Developed (Section 4)
Falsification	3 routes (math)	1 additional route (dark matter)

The papers are independent. Paper #82 stands without this one. This one requires Paper #82 for mathematical grounding.

6. Conclusion

The geometry is mathematics. The observer makes it physics. The distinction between the two IS the geometry.

Dark matter is not a mystery — it is the part of the spectrum that doesn't close into particles. The observer is not an add-on — the geometry forces observers as part of its self-description. The ur-axiom — one bit, one distinction — is the seed from which $\text{rank} = 2$ grows.

These claims are separable, falsifiable, and subordinate to the mathematics. The math (Paper #82) does not need the philosophy. But the philosophy needs the math.

References

[Paper #82] $1/\text{rank}$: Seven Famous Problems as One Geometric Invariant (this collaboration). [T944] Rank Forcing. [T1001] Observer as Conditional Expectation. [T1370] Information Completeness. [T1377] Self-Description Forcing. [T1431] Observer Instantiates Physics. [T1440] Consciousness = 50% (Toy 1440, 8/8). [Langlands 1976] On the Functional Equations Satisfied by Eisenstein Series. [DarkMatterCalculation.md] BST dark matter calculation — 175 galaxies, 0 parameters. [Toy 1637] Dark matter as incomplete windings on S^1 (9/9 PASS). $DM/\text{baryon} = r^4/N_c = 16/3$. [Toy 1642] Born rule from Bergman reproducing kernel (12/12 PASS). $K(z,z) = (1-|z|^2)^{-g}$.